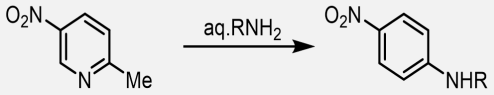
北京化工大学
Beijing University of Chemical Technology

N-C转换从吡啶合成苯

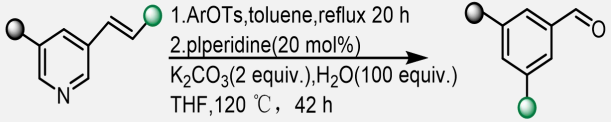
基地2301　　眭晨

背景介绍

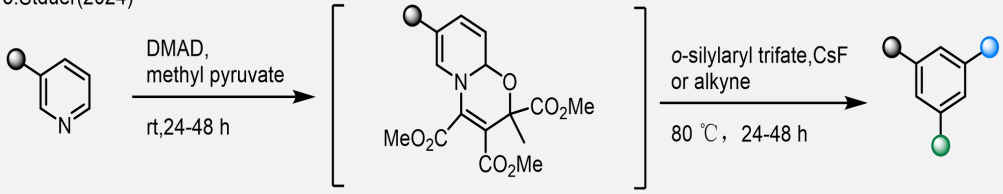
1.Kost-Sagitullin rearrangement(1978)



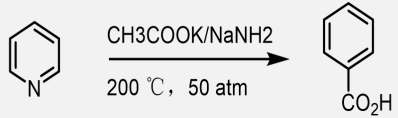
2.Kano and Morofull(2023)



3.Stduer(2024)



4.Toekelt(1964)



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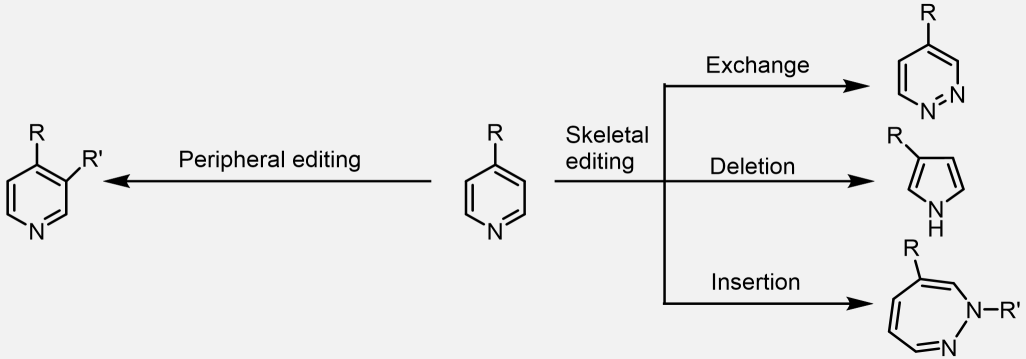

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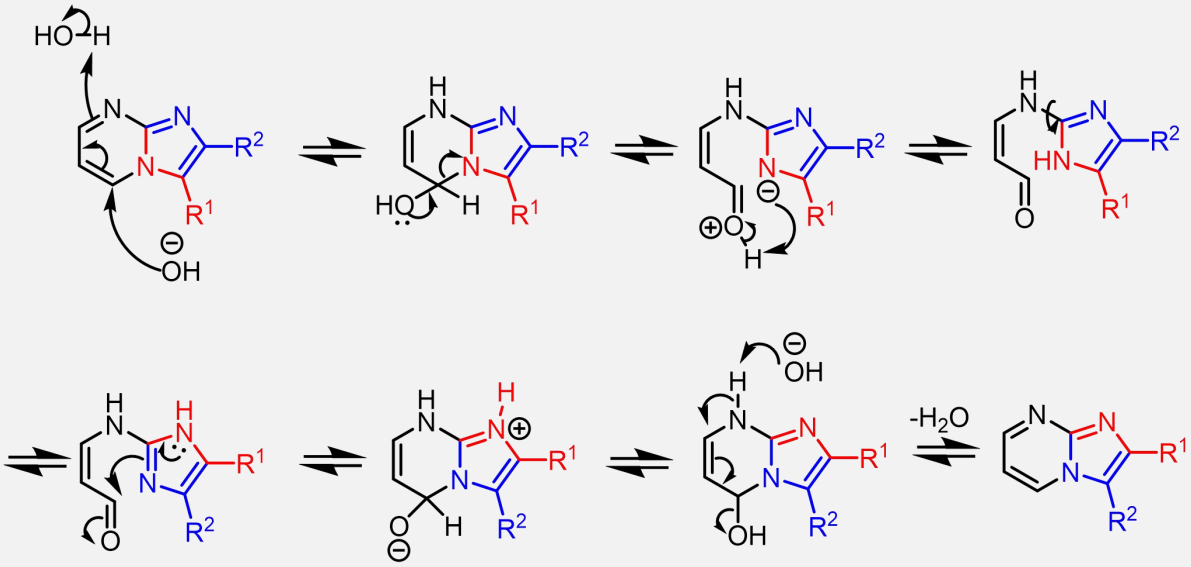
a.Zincke pyridinium反应

b.ANRORC反应

c.去芳构化环化、[4+2]环加成、芳构化逆环加成

ANRORC

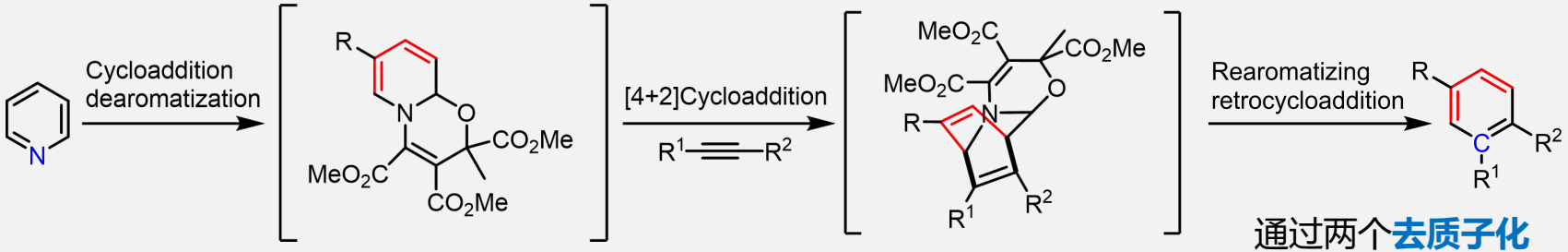


反应示意图

ANRORC缩写指的是Addition of the Nucleophile, Ring Opening, and Ring Closure

具体的过程包括：
(1) 亲核加成
(2) 开环
(3) 单键的旋转
(4) 关环
(5) 亲核物质的离去

计划方案



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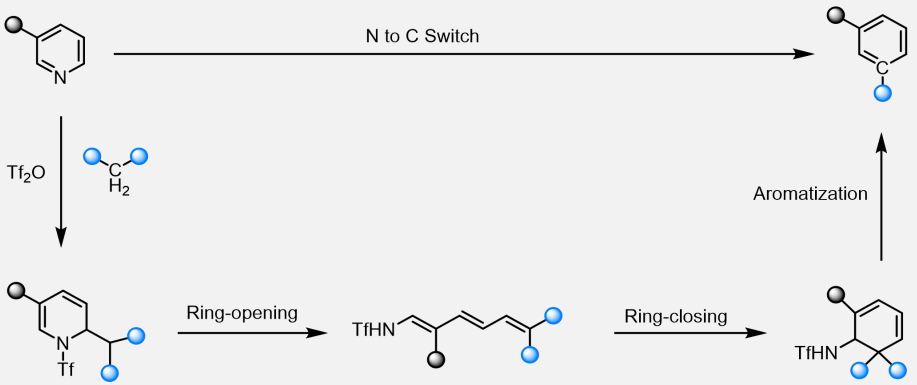
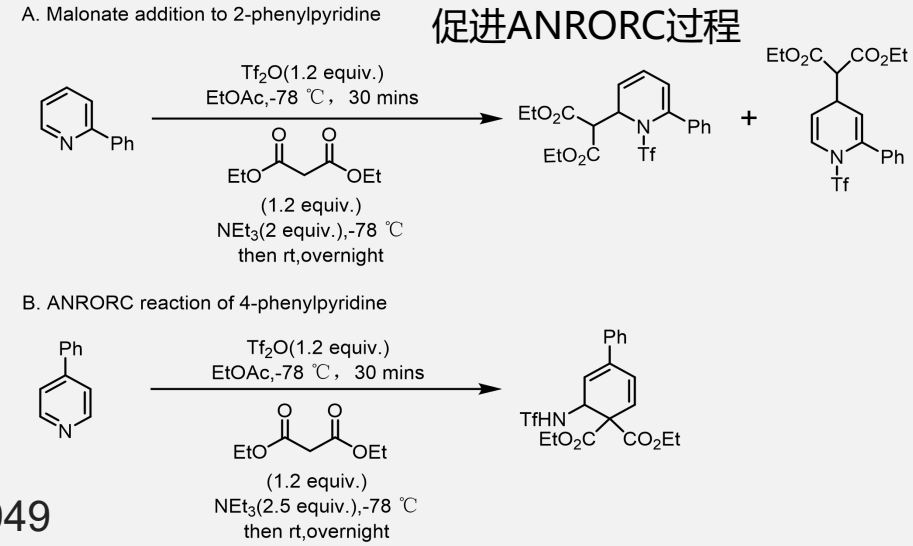
通过两个**去质子化**步骤，在分子间加入酸性亚甲基组分，促进ANRORC过程





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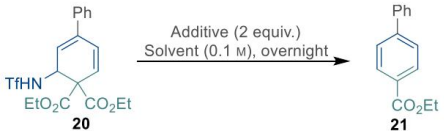
官能团适用范围与工艺优化

Table 1. Aromatization optimization

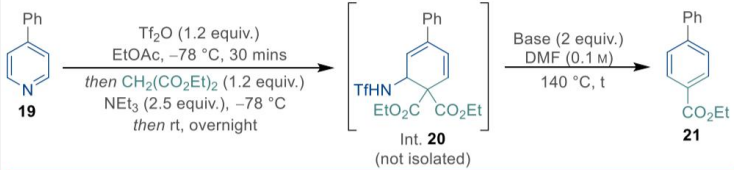


换溶液，DMF效果好

Entry	Solvent	Additive	Temp. (°C)	Conversion (%)[a]	Yield (%)[a]
1	EtOAc	NEt_3	80	15	2
2	EtOAc	DBU	80	66	9
3	Toluene	NEt_3	100	27	11
4	DMF	NEt_3	100	81	13
5	DMF	TsOH[b]	100	18	0
6	DMF	DBU	100	93	0
7	DMF	NEt_3	140	100	80
8	Mesitylene	NEt_3	140	67[c]	36[c]
9	DMF	K_2CO_3	140	100	79
10	DMF	DBU	140	100	69

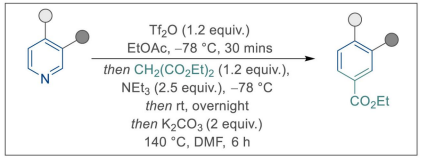
Table 2. Optimization of a one-pot protocol

换碱，K_2CO_3效果好

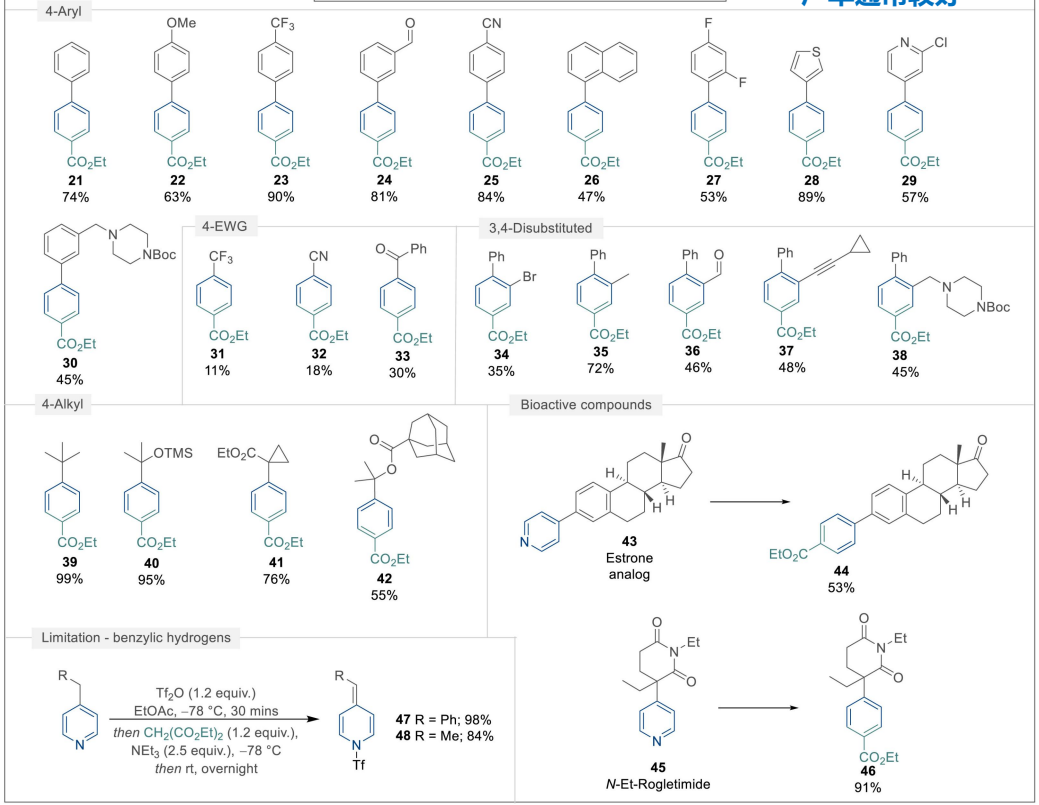


Entry	Base	Time (h)	Unreacted Int. 20 (%)[a]	Yield (%)[a]
1	NEt_3	3	0	50
2	N/A	3	33	38
3	DIPEA	3	0	60
4	Collidine	3	3	57
5	Cs_2CO_3	3	17	60
6	K_2CO_3	3	18	56
7	K_2CO_3	6	0	71

Pyridine Scope



官能团耐受性好
包括醛、叔胺取代、噻吩
杂二烯等
产率通常较好



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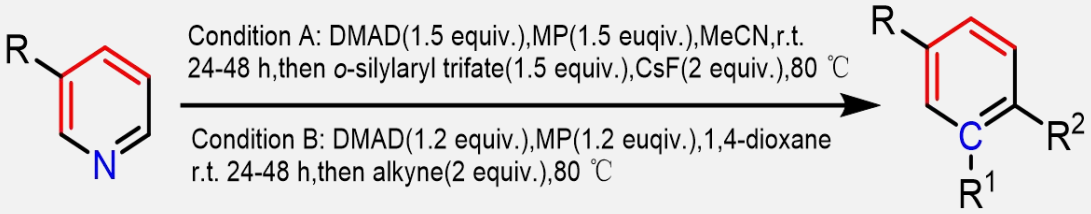
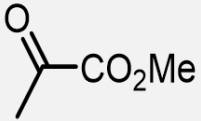
官能团适用范围与工艺优化



DMAF: MeO_2C—≡—CO_2Me

MP: 

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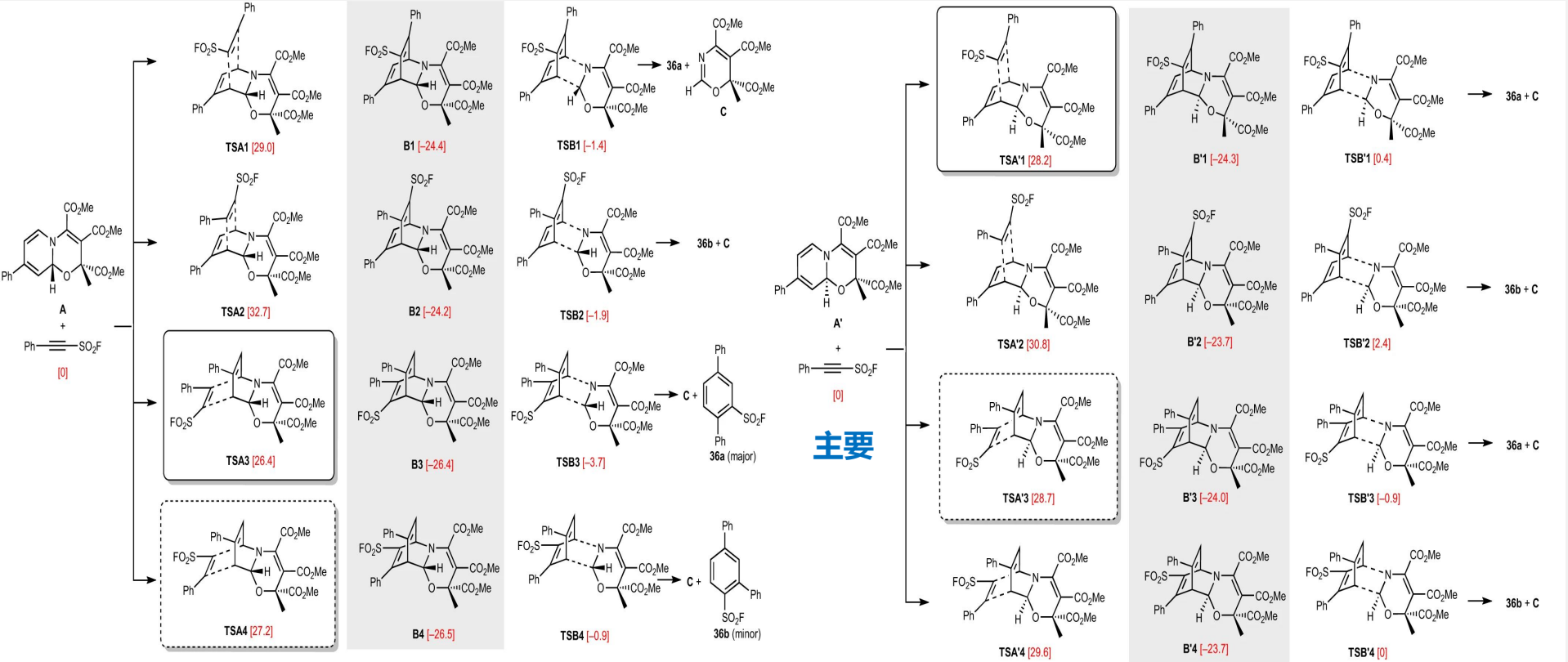
1）吡啶与乙炔二羧酸酯和丙酮酸酯的去芳构化环化反应可以在各种溶剂中高效进行，从而获得oxazino-吡啶中间体

2）oxazino-吡啶中间体与原位生成的芳炔在乙腈，或者与炔烃在1,4-二恶烷/甲苯于80℃反应便可获得取代的苯和萘衍生物，而且温度过高或者过低均会降低产率。

3）在最优条件下，作者对吡啶的底物范围进行了考察，结果显示酯基、氰基、苯氧基、卤素原子、烯基、甲基、吡唑、苯并呋喃、炔基、羰基、吡啶、吲哚、噻吩取代的吡啶均能兼容该反应，以中等至良好的产率获得所需产物。

4）邻位取代吡啶在去芳构化和随后的[4+2]环加成反应中都增加了空间位阻，进而导致产率降低；而间位取代吡啶在初始去芳构化步骤的区域选择性转移到最终产物，进而产生具有两种组成异构体的萘。其次，含有氟、烷基和O-取代基的对称二取代芳炔、不对称芳炔甚至活化的炔烃也能顺利地转化为相应产物，但是未活化炔烃或末端炔烃却无法进行该反应。

机理研究



syn-加成有利　　TSB3,TSB4能垒较低　　能垒都较低

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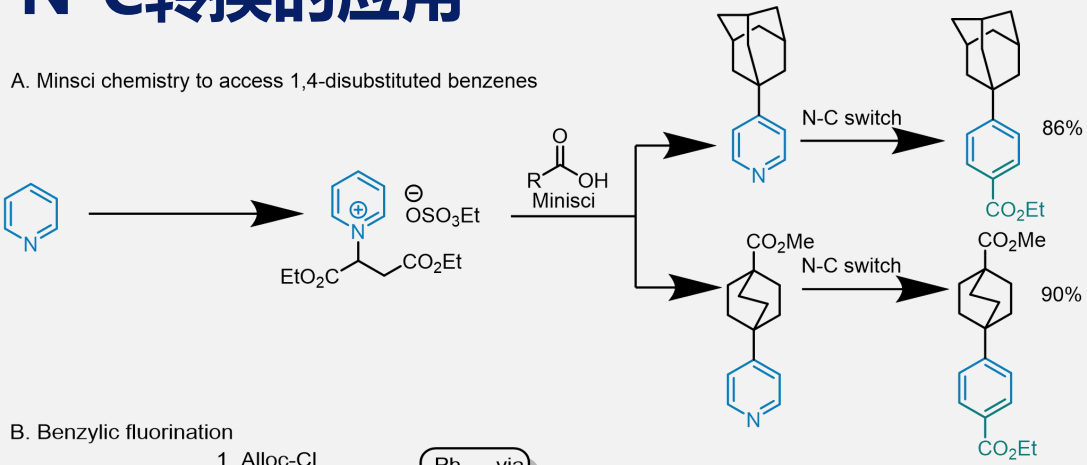
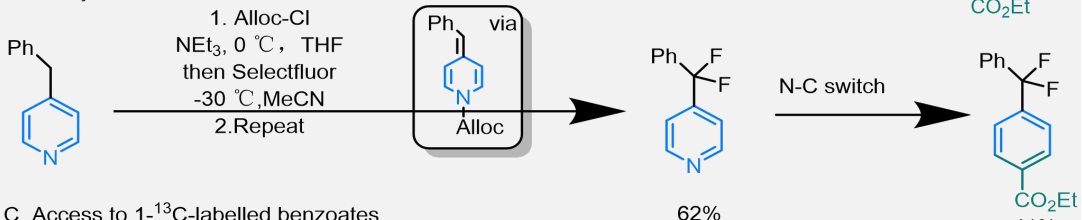
N-C转换的应用

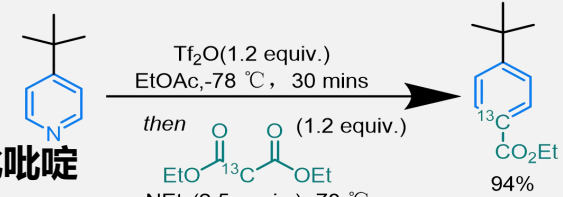
A. Minsci chemistry to access 1,4-disubstituted benzenes



B. Benzylic fluorination



C. Access to 1-^{13}C-labelled benzoates



A.三级烷基化吡啶
B.苄基氟化
C.同位素标记

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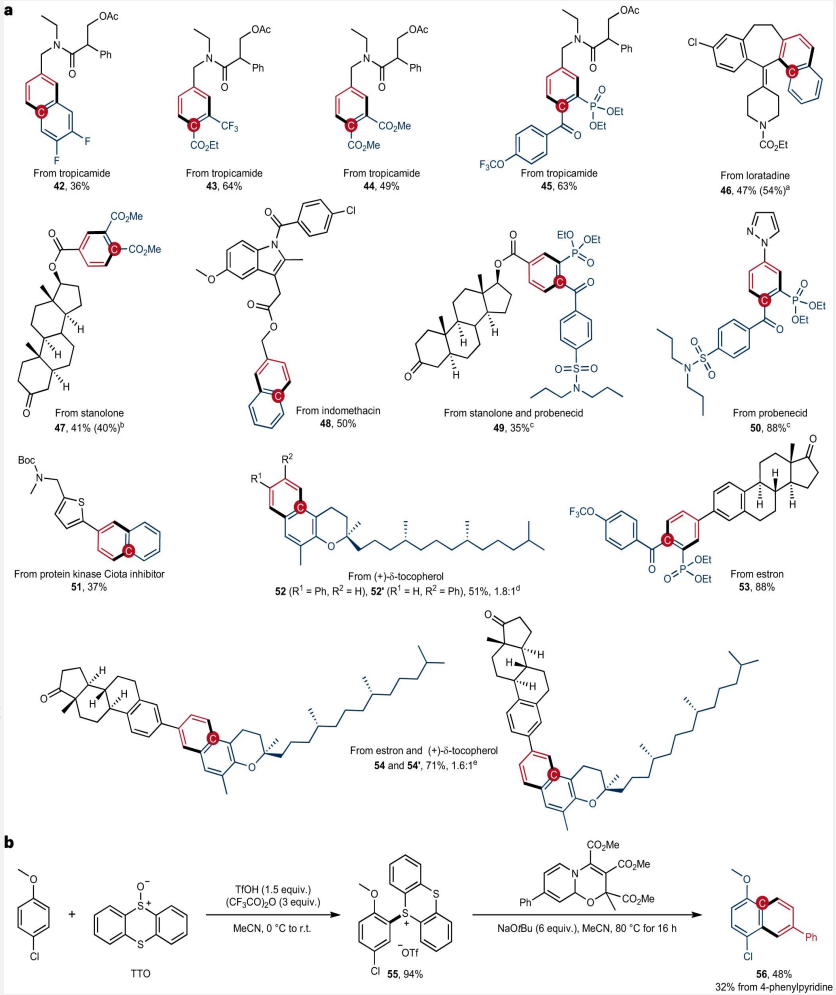

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请各位老师批评指正

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